Field Shift: Adapting Farms with NASA Data

Data	Source
Rainfall	GPM / IMERG
Soil moisture	SMAP
Temperature, humidity, radiation	POWER / MERRA-2
Vegetation health (NDVI/EVI)	MODIS, Landsat, HLS
Evapotranspiration	MODIS / ECOSTRESS
Land cover / crop history	Landsat, MODIS land cover
Flood/drought indicators	GRACE, GPM

Possible Training Features

Possible Inputs

Important

- Soil info
- Weather
- Vegetation health and seasonal growth
- Evapotranspiration and water demand
- Farmer priorities and irrigation access
- Crop prices and profitability

Additional

- Land cover and land-cover change
- Land-surface temperature
- Plant water stress
- Elevation and terrain
- Surface water and flood-related observations

Possible Outputs / System Recommendations

Crop-rotation sequence

Recommend a sequence of crops across successive growing seasons, rather than selecting a crop in isolation.

Soil-health improvement

Explain how candidate rotations may affect nutrient balance, nitrogen fixation, organic matter inputs, and other relevant soil-health indicators.

Water-smart crop selection

Compare candidate rotations by estimated seasonal water requirements, rainfall availability, soil moisture, and irrigation constraints.

Climate-risk-aware planning

Identify rotations that may be more resilient to heat, dry spells, excessive rainfall, or other locally relevant hazards.

Trade-off comparison

Give farmers several viable alternatives, such as the best balanced option, a water-conservation option, and a soil-health-focused option.